Ethics under uncertainty

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Introduction

Many decisions involve *uncertainty* about the consequences of actions.

For someone who wants to be *ethical*, there is another source of uncertainty: uncertainty about the correct ethical theory.

This lecture will discuss tools for dealing with both kinds of uncertainty.

Outline

I. Decisions under uncertainty

II. Metaethics crash course

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I . Decisions under uncertainty

Consider an agent who is presented with a set $\mathcal{A}$ of possible *actions*.

Each action will yield a consequence from a set $\mathcal{X}$ of possible *outcomes*.

The agent wants to choose the "best" action.

Problem. The agent is not sure what outcome will result from each action.

There are three ways to model this:

1. **von Neumann-Morgenstern** (vNM) **framework**. Each action determines a probability distribution ("lottery") over $\mathcal{X}$.

Let $\Delta(\mathcal{X})$ denote the set of such lotteries. Goal: Pick best $P \in \Delta(\mathcal{X})$.

2. **Savage framework**. There is a space $\mathcal{S}$ of possible "states of nature". The true state is unknown.

An action is a function $\alpha : \mathcal{S} \longrightarrow \mathcal{X}$.

If you choose action α, and state turns out to be s, then you will get $\alpha(s)$.

3. **Anscombe-Aumann** (AA) **framework.** (Combine Savage with vNM) An action is a function $\alpha : \mathcal{S} \longrightarrow \Delta(\mathcal{X})$.

If you choose α, and state is s, then you will face the lottery $\alpha(s) \in \Delta(\mathcal{X})$.

Expected utility

To make rational decisions, we need a *preference order* $\succcurlyeq$ on $\mathcal{A}$.

Question. How should we construct this preference order?

Gold standard. Maximize expected utility.

In the vNM framework, this means that there is a *utility function* $u : \mathcal{X} \longrightarrow \mathbb{R}$ such that, for any lotteries $P, Q \in \Delta(\mathcal{X})$,

$$P \succcurlyeq Q \quad \iff \quad \overline{u}(P) \geq \overline{u}(Q), \qquad (*)$$

where $\overline{u}(P) := \int_{\mathcal{X}} u(x) \, \mathrm{d}P[x]$ etc. (If $\mathcal{X}$ is finite, then $\overline{u}(P) := \sum_{x \in \mathcal{X}} P(x)\, u(x)$.)

vNM Theorem. *$\succcurlyeq$ satisfies three normatively appealing axioms if and only if there is some $u : \mathcal{X} \longrightarrow \mathbb{R}$ such that $\succcurlyeq$ is represented as in $(*)$.*

Furthermore, in this representation, u is unique up to positive affine transformations (P.A.T.).

In other words: if $v : \mathcal{X} \longrightarrow \mathbb{R}$ is another function that yields a representation $(*)$, then there exist constants $r \in \mathbb{R}$ and $s > 0$ such that $v = s\, u + r$.

The vNM Independence axiom

Here, "*axiom*" does *not* mean "obviously true statement" (as in Euclidean geometry or ZF set theory) or "defining condition" (as in group theory).

Rather, it means "*desideratum*": a property that you would *like to* satisfy.

The key axiom in the vNM Theorem is **Independence**:

For all $P_1, P_2 \in \Delta(\mathcal{X})$, if $P_1 \succcurlyeq P_2$, then for all $Q \in \Delta(\mathcal{X})$ and $t \in [0,1]$,

$$t\,P_1 + (1-t)\,Q \;\succcurlyeq\; t\,P_2 + (1-t)\,Q.$$

Idea. These are *compound lotteries*, involving a biased coin that lands "tails" with probability t and "heads" with probability $1-t$.

If the coin lands "heads", then you will get Q with either choice.

So the choice should be determined by what happens when it lands "tails".

So, if you preferred P_1 over P_2, ...
.... then you should prefer $t\,P_1 + (1-t)\,Q$ over $t\,P_2 + (1-t)\,Q$.

Savage expected utility

In the Savage Framework, an EU representation has two components:

- a utility function $u : \mathcal{X} \longrightarrow \mathbb{R}$; and
- a probability measure ρ on $\mathcal{S}$,

such that, for any actions $\alpha, \beta \in \mathcal{A}$,

$$\alpha \succcurlyeq \beta \quad \Longleftrightarrow \quad \mathbb{E}_\rho[u \circ \alpha] \geq \mathbb{E}_\rho[u \circ \beta], \qquad (*)$$

where $\mathbb{E}_\rho[u \circ \alpha] := \int_{\mathcal{S}} u[\alpha(s)]\ \mathrm{d}\rho[s]$, etc.

Idea. ρ encodes the agent's *beliefs*.

u encodes the agent's *desires*.

Savage Theorem. *$\succcurlyeq$ satisfies six normatively appealing axioms if and only if it has an expected utility representation $(*)$.*

Furthermore, in this representation the probability measure ρ is unique, and the utility function u is unique up to P.A.T.

The Sure-Thing Principle

The key axiom in Savage's Theorem is the **Sure-Thing Principle** (STP)

For any subset $\mathcal{R} \subseteq \mathcal{S}$, let $\mathcal{A}_{\mathcal{R}} := \{\alpha_{|\mathcal{R}};\ \ \alpha \in \mathcal{A}\}$.

The **Sure-Thing Principle** says:

For any measurable subset $\mathcal{R} \subseteq \mathcal{S}$, there is a "conditional" preference order $\succcurlyeq_{\mathcal{R}}$ on $\mathcal{A}_{\mathcal{R}}$ such that: for any $\alpha, \beta \in \mathcal{A}$ with $\alpha(s) = \beta(s)$ for all $s \in \mathcal{S} \setminus \mathcal{R}$, we have

$$\alpha \succcurlyeq \beta \quad \Longleftrightarrow \quad \alpha_{|\mathcal{R}} \succcurlyeq_{\mathcal{R}} \beta_{|\mathcal{R}}.$$

The interpretation is very similar to the vNM Independence axiom.

If the true state is in $\mathcal{S} \setminus \mathcal{R}$, then α and β deliver identical outcomes.

So in this case, it doesn't matter which one you choose.

Thus, the choice between α and β should be entirely determined by comparing $\alpha_{|\mathcal{R}}$ with $\beta_{|\mathcal{R}}$.

Expected utility in the AA world

In the Anscombe-Aumann framework, an expected utility representation has the same two components as in Savage.

Now, for any (lottery-valued) actions $\alpha, \beta \in \mathcal{A}$, we require:

$$\alpha \succcurlyeq \beta \quad \Longleftrightarrow \quad \mathbb{E}_\rho[\overline{u} \circ \alpha] \geq \mathbb{E}_\rho[\overline{u} \circ \beta], \qquad (*)$$

where $\mathbb{E}_\rho[\overline{u} \circ \alpha] := \int_{\mathcal{S}} \overline{u}[\alpha(s)]\ \mathrm{d}\rho[s] = \int_{\mathcal{S}} \int_{\mathcal{X}} u(x)\ \mathrm{d}\alpha(s)[x]\ \mathrm{d}\rho[s]$ etc.

AA Theorem. *$\succcurlyeq$ satisfies three normatively appealing axioms if and only if it has an expected utility representation $(*)$.*

In this representation ρ is unique, and u is unique up to P.A.T.

The AA Independence axiom

The key axiom in the AA Theorem is a version of vNM's *Independence*.

To state this, recall that AA actions take values in $\Delta(\mathcal{X})$.

So we can form convex combinations of AA actions, just as we form convex combinations of lotteries.

The AA *Independence* axiom says:

For all $\alpha_1, \alpha_2 \in \mathcal{A}$, if $\alpha_1 \succcurlyeq \alpha_2$, then for any $\beta \in \mathcal{A}$ and any $t \in [0,1]$,

$$t\,\alpha_1 + (1-t)\,\beta \;\succcurlyeq\; t\,\alpha_2 + (1-t)\,\beta.$$

The rationale is the same as that of the vNM *Independence* axiom.

The problem of the prior

Problem. Where does ρ come from?

In the vNM model, all probabilities are known and "objective".

But in the Savage/AA models, ρ is not specified in advance.

It is *derived* as part of the representation.

Savage and AA just say such a ρ *exists*: they don't tell you how to find it.

In the standard view, ρ just represents the "subjective beliefs" of the agent.

It might not correspond to anything in objective reality.

It is completely arbitrary.

For this reason, Savage and AA are often called *subjective* expected utility (SEU) representations.

The problem of the prior

However, in decisions with very high stakes (especially ethical stakes), it doesn't seem satisfactory to just use "arbitrary" or "subjective" beliefs.

Furthermore, in situations of *ambiguity*, where we are radically ignorant about the nature of reality, positing a precise probability measure over $\mathcal{S}$ is not only arbitrary; it is hubristic.

It is more intellectually honest to *admit* that we do not know enough to specify a precise probability measure, and instead use some other criterion.

This has lead to a rich theory of *decisions under ambiguity*, with various non-SEU models of decision-making.

Many of these replace the *unique* probability measure of Savage/AA with a *set* of probability measures.

These are called *multiple priors* models.....

Minimum EU preferences

As usual, $\mathcal{A}$ is a set of functions from $\mathcal{S}$ to $\mathcal{X}$ (or $\Delta(\mathcal{X})$).

$\succcurlyeq$ is a preference order on $\mathcal{A}$.

We say that $\succcurlyeq$ is a *minimum EU* (MEU) *preference* if there is a compact, convex subset $\mathcal{P} \subseteq \Delta(\mathcal{S})$ and a function $u : \mathcal{X} \longrightarrow \mathbb{R}$ such that, for all $\alpha, \beta \in \mathcal{A}$,

$$\alpha \succcurlyeq \beta \quad \Longleftrightarrow \quad \min_{\rho \in \mathcal{P}} \mathbb{E}_\rho[u \circ \alpha] \geq \min_{\rho \in \mathcal{P}} \mathbb{E}_\rho[u \circ \beta],$$

Interpretation.
$\mathcal{P}$ is the set of probability measures that the agent thinks are "possible".

The agent is extremely pessimistic.

So she evaluates each act by its *minimum* EU over all possible probability measures in $\mathcal{P}$.

Remark. if $\mathcal{P}$ is a singleton, then MEU reduces to SEU.

Hurwicz preferences

Arguably, MEU is too pessimistic.

We say that $\succcurlyeq$ is a *Hurwicz preference* if there is a compact, convex subset $\mathcal{P} \subseteq \Delta(\mathcal{S})$, a function $u : \mathcal{X} \longrightarrow \mathbb{R}$ and a constant $h \in [0, 1]$ such that, for all $\alpha, \beta \in \mathcal{A}$,

$$\alpha \succcurlyeq \beta \quad \iff \quad V_h(\alpha;\, u, \mathcal{P}) \geq V_h(\beta;\, u, \mathcal{P}),$$

where $V_h(\alpha;\, u, \mathcal{P}) := h \min_{\rho \in \mathcal{P}} \mathbb{E}_\rho[u \circ \alpha] + (1 - h) \max_{\rho \in \mathcal{P}} \mathbb{E}_\rho[u \circ \alpha]$.

Interpretation.
$\mathcal{P}$ is the set of probability measures that the agent thinks are "possible".

h is an index of "pessimism".

$h = 1$ yields MEU preferences.

But lower values of h cause the agent to think more about the *maximum* possible EU as well as the *minimum* EU.

SOSEU preferences

Problem. Both MEU and Hurwicz focus exclusively on extremes.

We say that $\succcurlyeq$ is a *second-order subjective expected utility* (SOSEU) *preference* if there is a "second-order probability measure" $\mu \in \Delta[\Delta(\mathcal{S})]$, a function $u : \mathcal{X} \longrightarrow \mathbb{R}$ and an increasing (usually concave) function $\phi : \mathbb{R} \longrightarrow \mathbb{R}$ such that for all $\alpha, \beta \in \mathcal{A}$,

$$\alpha \succcurlyeq \beta \quad \iff \quad \int_{\Delta(\mathcal{S})} \phi\Big(\mathbb{E}_\rho[u \circ \alpha]\Big)\, \mathrm{d}\mu[\rho] \geq \int_{\Delta(\mathcal{S})} \phi\Big(\mathbb{E}_\rho[u \circ \beta]\Big)\, \mathrm{d}\mu[\rho].$$

Interpretation. μ describes the agent's "second-order beliefs" about the likelihood that each element of $\Delta(\mathcal{S})$ is the "true" probability measure.

ϕ captures the agent's *ambiguity aversion*: the extra discomfort that she feels about actions that involve unknown probability distributions.

Note. If ϕ is linear, then SOSEU reduces to SEU.

So it is only an interesting model if ϕ is nonlinear.

Variational preferences

We say that $\succcurlyeq$ is a *variational preference* if there is a convex, lower semicontinuous function $\Gamma : \Delta(\mathcal{S}) \longrightarrow [0, \infty]$ with minimum value zero, and a function $u : \mathcal{X} \longrightarrow \mathbb{R}$ and such that for all $\alpha, \beta \in \mathcal{A}$,

$$\alpha \succcurlyeq \beta \quad \iff \quad \min_{\rho \in \Delta(\mathcal{S})} \Big(\mathbb{E}_\rho[u \circ \alpha] + \Gamma(\rho) \Big) \geq \min_{\rho \in \Delta(\mathcal{S})} \Big(\mathbb{E}_\rho[u \circ \beta] + \Gamma(\rho) \Big).$$

Interpretation. Γ assigns more "weight" or "relevance" to some probability measures over $\mathcal{S}$ than to others.

$\Gamma(\rho_1) < \Gamma(\rho_2) \implies \rho_1$ is more relevant than ρ_2.

In particular, $\Gamma(\rho) = \infty \implies \rho$ is totally irrelevant.

Example. Let $\mathcal{P} \subseteq \Delta(\mathcal{S})$ be compact and convex. Define Γ by:

- $\Gamma(\rho) = 0$ if $\rho \in \mathcal{P}$; and
- $\Gamma(\rho) = \infty$ if $\rho \notin \mathcal{P}$.

Then Γ is convex and lower semicontinuous, and the variational preferences induced by Γ are equivalent to the MEU preferences induced by $\mathcal{P}$.

Idea: Variational preference can be seen as a "fuzzy set" version of MEU.

Bewley preferences

Let $\succcurlyeq$ be a *preorder* on $\mathcal{A}$ (i.e. a binary relation that is transitive and reflexive, but possibly incomplete).

We say that $\succcurlyeq$ is a *Bewley preference* if there is a set $\mathcal{P} \subseteq \Delta(\mathcal{S})$ and a function $u : \mathcal{X} \longrightarrow \mathbb{R}$ such that, for all $\alpha, \beta \in \mathcal{A}$,

$$\alpha \succcurlyeq \beta \quad \iff \quad \mathbb{E}_\rho[u \circ \alpha] \geq \mathbb{E}_\rho[u \circ \beta], \quad \text{for all } \rho \in \mathcal{P}.$$

Idea. Bewley is committed to EU-maximization....

But it takes very seriously the fact that the agent *has no idea* what is the true probability measure over $\mathcal{S}$, except that it is some element of $\mathcal{P}$.

Other decision rules either commit us to extreme *pessimism* (MEU, variational), or they involve extra parameters (e.g. h, μ, ϕ) that might be just as difficult to identify as ρ itself.

Bewley says: if you don't know the true probability measure, then just *admit* that you don't know, and reserve judgement except in situations when *all* elements of $\mathcal{P}$ tell you that α is better than β.

For each of the aforementioned non-SEU approaches (MEU, Hurwicz, SOSEU, variational, Bewley), there are *axiomatic characterization* theorems of the form:

Theorem. $\succcurlyeq$ *satisfies normative axioms* A, B, C, *and* D *if and only if it has a representation of type* X.

Furthermore, in this representation, the gadget encoding "beliefs" (e.g. the set $\mathcal{P}$*, the second-order probability* μ*) is unique, while the gadget encoding "utilities" (e.g.* u *or* $\mathcal{U}$*) is unique up to P.A.T.*

In most cases, a key axiom is a carefully *weakened* version of Savage's Sure-Thing Principle or the AA Independence axiom.

(Major contributers include: Tzachi Gilboa, David Schmeidler, Peter Wakker, Peter Klibanoff, Massimo Marinacci, Sujoy Mukerjee, Fabio Maccheroni, etc.)

II. Metaethics crash course

Three kinds of moral theories (21/40)

Moral philosophical theories can be divided into three families:

- **Consequentialism:** the locus of moral judgement is *consequences*;
- **Deontology:** the locus of moral judgement is *actions*;
- **Virtue ethics:** the locus of moral judgement is *character*.

Consequentialism. Consequentialist theories judge actions by their *consequences*.

Action α is better than action β if α yields better *long-term consequences* than β, all things considered, for everyone involved.

(In situations of uncertainty, this can be amended to "better *expected* consequences".)

Utilitarianism is the prototypical consequentialist theory.

Likewise, economists judge policies by the *social welfare* they generate. This is a consequentialist approach.

Three kinds of moral theories (22/40)

Deontology. Deontological theories say that actions are intrinsically right or wrong, *independent* of their consequences.

Kant's Categorical Imperative is the prototypical deontological theory.

The moral codes of most religions are a list of rules: "Thou shalt not X".

So are most legal systems. These are also deontological theories.

Virtue Ethics. Instead of good consequences or good actions, virtue ethics says you should aim for a good *character*.

Virtue ethicists say that we should cultivate "virtues" such as *patience, honesty, courage, compassion, temperance*, etc.

Many ancient ethical systems (Aristotelian ethics, Stoicism, Confucianism, Buddhism) have this form.

Also, imitating a role model ("What would Gandhi do?") is a kind of virtue ethics.

Realism vs. antirealism

Another important distinction is between moral *realism* and *antirealism*.

Moral *realists* believe there are *objective, universal moral facts*, the same way that there are objective, universal facts in science and mathematics.

The goal of moral philosophy is to *discover* these facts, just as the goal of mathematics is to discover mathematical facts.

Moral *antirealists* say that there are no objective moral facts.

There are several versions of antirealism. Here are a few....

- *Constructivism*: Moral rules are just the *conventions* of a society.

  Like the norms of social etiquette, or the grammar of a language, or the rules of a ball game, they are true for a particular society in a particular time or place, but not "universal" or "objective".

- *Evolutionary debunking*: Moral intuitions result from natural selection, like parental instincts, in-group bias, or a taste for sugar.

  They were selectively advantageous for small bands of hominids.

- *Noncognitivism*: Moral language doesn't make factual assertions at all.

  Instead, it expresses emotions of approval or disapproval ("Yay!", "Boo!") or issues prescriptions ("Do not steal.").

The idea of "moral uncertainty" seems to assume moral *realism*.

But it is also applicable for *some* kinds of moral antirealism, such as constructivism.

III. Moral Uncertainty

Moral uncertainty

Classical decision theory is only concerned with rationality, not morality.

But suppose that we don't just want to make a *rational* choice —we want to make the *right* choice.

Problem. There are many different moral theories.

We don't know which one is correct.

Idea. Let $\mathcal{T}$ be a set of moral theories.

Let $\mathcal{A}$ be a set of actions.

Any action α defines a function $\overline{\alpha} : \mathcal{T} \longrightarrow \mathbb{R}$, where, for all $t \in \mathcal{T}$, $\overline{\alpha}(t)$ is the *choiceworthiness* of α, according to t.

Let ρ be a probability measure on $\mathcal{T}$, describing how much credence the agent assigns to each moral theory.

Question. How should we choose, so as to be "as moral as possible"?

My Favourite Theory

Idea. ("My Favourite Theory")

Let $t^* \in \mathcal{T}$ be the theory with maximal ρ-probability.

Choose the action $\alpha \in \mathcal{A}$ that maximizes $\overline{\alpha}(t^*)$.

Formally, for all $\alpha, \beta \in \mathcal{A}$,

$$\alpha \succcurlyeq \beta \iff \overline{\alpha}(t^*) \geq \overline{\beta}(t^*).$$

Problems. 1. If $\mathcal{T}$ is uncountably infinite and ρ is a nonatomic probability measure, then there might not be any such t^*.

2. Even if there is such a t^*, it might not be unique.

3. Even if there is a unique t^*, "My Favourite Theory" runs into a lot of problems and paradoxes, and is generally rejected by moral philosophers.

Expected choiceworthiness

Better idea. *Maximize expected choiceworthiness.*

Formally, for all $\alpha, \beta \in \mathcal{A}$,

$$\alpha \succcurlyeq \beta \quad \iff \quad \mathbb{E}_\rho[\overline{\alpha}] \geq \mathbb{E}_\rho[\overline{\beta}].$$

where $\mathbb{E}_\rho[\overline{\alpha}] := \int_{\mathcal{T}} \overline{\alpha}(t)\ \mathrm{d}\rho[t]$.

This has been advocated by Kryster Bykvist, Will McAskill, Toby Ord, etc.

Problem. How are we supposed to find the credence measure ρ over $\mathcal{T}$?

Idea. Clearly, we can combine this basic idea with any of the non-SEU decision rules that were described earlier.

For example, we could introduce a *set* $\mathcal{P} \subseteq \Delta(\mathcal{T})$ and a concave increasing transformation $u : \mathbb{R} \longrightarrow \mathbb{R}$, and then introduce a Bewley-type criterion:

$$\alpha \succcurlyeq \beta \iff \mathbb{E}_\rho[u \circ \overline{\alpha}] \geq \mathbb{E}_\rho[u \circ \overline{\beta}], \quad \text{for all } \rho \in \mathcal{P}.$$

Twofold uncertainty

We can simultaneously cope with uncertainty about the correct moral theory *and* uncertainty about the true state of nature.

As before, let $\mathcal{S}$ be a set of states of nature.

Regard each action α as defining a function $\overline{\alpha} : \mathcal{S} \times \mathcal{T} \longrightarrow \mathbb{R}$.

Here, $\overline{\alpha}(s,t)$ is the choiceworthiness (according to theory t) of the action α if the state of nature s obtains.

For instance: let $\mathcal{X}$ be a set of outcomes, and suppose (as with Savage) that actions are functions from $\mathcal{S}$ to $\mathcal{X}$.

A moral theory t is *purely consequentialist* if there is a function $v_t : \mathcal{X} \longrightarrow \mathbb{R}$ (the *axiology* of theory t) such that $\overline{\alpha}(s,t) = v_t[\alpha(s)]$ for all $s \in \mathcal{S}$.

In other words: the choiceworthiness of α depends *only* on its consequences. (Example: utilitarianism.)

A moral theory t is *purely deontological* if $\overline{\alpha}(s,t)$ is *independent* of s.

In other words: actions are judged to be "right" or "wrong", independent of their consequences. (Example: Kantianism.)

Twofold uncertainty

This framework is flexible enough to accommodate both purely consequentialist and purely deontological theories, as well as theories that lie on the continuum between these extremes. (Virtue ethics? Unclear.)

We can generalize the expected-choiceworthiness criterion to this setting.

Let $\rho \in \Delta(\mathcal{S} \times \mathcal{T})$, and stipulate that

$$\alpha \succcurlyeq \beta \quad \Longleftrightarrow \quad \mathbb{E}_\rho[\overline{\alpha}] \geq \mathbb{E}_\rho[\overline{\beta}].$$

where now, $\mathbb{E}_\rho[\overline{\alpha}] := \int_{\mathcal{S}\times\mathcal{T}} \overline{\alpha}(s,t)\ \mathrm{d}\rho[s,t]$.

(**Note.** We do *not* assume that ρ is a product measure.
In other words, we do not assume that the state of nature and the true moral theory are probabilistically independent.)

Again, we can extend this any of the non-SEU decision rules described earlier (e.g. MEU, Hurwicz, SOSEU, Bewley) in the obvious way.

Ordinal noncomparable choiceworthiness

Problem.
Expected-choiceworthiness (and its non-SEU variants) assume that "choiceworthiness" is *cardinal* and *comparable across theories*.

In other words, for any moral theory t, the choiceworthiness of an action α is a *real number* $\overline{\alpha}(t)$.

Furthermore, given two theories t_1 and t_2, we can compare $\overline{\alpha}(t_1)$ and $\overline{\alpha}(t_2)$ as if they are measured on the same "scale".

Neither of these assumptions is obviously true.

Suppose instead that choiceworthiness is *ordinal*, and *not* comparable across theories.

Formally, for all $t \in \mathcal{T}$, there is a weak order $\succcurlyeq_t$ on $\mathcal{A}$ describing *ordinal choiceworthiness according to theory* t.

But we cannot say, "α is more choiceworthy for t_1 than β is for t_2."

Question. How can we handle moral uncertainty in this setting?

Moral uncertainty as social choice

Idea. This problem is analogous to classical Arrovian social choice theory.

Treat the moral theories in $\mathcal{T}$ as "voters" (weighted by probability).

Apply an Arrovian preference aggregation rule Φ to aggregate the profile of preferences $(\succcurlyeq_t)_{t\in\mathcal{T}}$ into some consensus preference $\succcurlyeq_* := \Phi[(\succcurlyeq_t)_{t\in\mathcal{T}}]$.

Bykvist, McAskill, and Ord argue that the *Borda rule* is the best choice.

Example. Suppose $\mathcal{A} = \{a, b, c, d\}$ and $\mathcal{T} = \{t_1, t_2, t_3\}$ (all equally likely).

Suppose $a \succ_1 b \succ_1 c \succ_1 d$, $\quad b \succ_2 a \succ_2 c \succ_2 d$, and $c \succ_3 d \succ_3 a \succ_3 b$.

The three theories assign *Borda scores* to the alternatives as follows:

	a	b	c	d
t_1	4	3	2	1
t_2	3	4	2	1
t_3	2	1	4	3
Total	**9**	8	8	5

Alternative a has the highest score. So we should choose a.

Sometimes choiceworthiness is not expressed as a weak order.

Some moral theories (e.g. deontological theories) make a binary distinction: either an action is *permissible*, or it is *forbidden*.

Beyond this, they do not deem some actions "better" than others.

Example. Suppose $\mathcal{A} = \{A, B, C, D, E\}$ and $\mathcal{T} = \{V, W, X, Y, Z\}$, with credences and permissibility as in the following table:

		Action				
Theory	Credence	*A*	*B*	*C*	*D*	*E*
V	10%	✗	✓	✓	✓	✓
W	10%	✗	✗	✓	✗	✓
X	15%	✗	✓	✗	✗	✓
Y	35%	✓	✗	✗	✓	✗
Z	30%	✓	✓	✗	✗	✗
Total		65%	55%	20%	45%	35%

Idea. Choose action permitted by the *largest total credence* of theories. In this case, it is A (permitted by Theories Y and Z, with 65% credence).

This is an application of the Approval Voting rule (Brams & Fishburn 1983).

IV. Problems

Problems with the social choice model

We have just considered *Arrovian preference aggregation* to resolve moral uncertainty when choiceworthiness is ordinal and noncomparable.

However, there are literally dozens of Arrovian preference aggregation rules, all with different strengths and weaknesses.

A panoply of impossibility theorems says that there is no "perfect" preference aggregation rule.

The choice seems somewhat arbitrary.

So this is not a very intellectually satisfactory approach.

Information and dynamic consistency

We have considered various non-SEU rules for decisions with ambiguity.

Now suppose the agent acquires new information.

How should she update her beliefs?

For SEU agents, the answer is clear: apply *Bayes rule*.

But for non-SEU agents, the correct generalization of Bayes is not obvious.

Also, updating beliefs can lead to perverse phenomena for non-SEU agents.

For example, learning new information can make the agent *worse off*.

So agents may exhibit *information aversion* (avoiding new information).

Also, in *intertemporal* decision problem, where an agent must make a series of choices over time, non-SEU agents can exhibit *dynamic inconsistency* (i.e. inability to make a plan and stick to it).

Indeed, Peter Hammond and Paolo Ghirardato have shown that EU maximization is the *only* dynamically consistent decision rule.

Unawareness

We have seen several models of decision-making with uncertainty.

But all of these models assume that the agent *knows* the state space $\mathcal{S}$ and the outcome space $\mathcal{X}$.

(And for moral uncertainty: she *knows* the full set $\mathcal{T}$ of moral theories.)

In some cases, this is unrealistic.

The agent may learn about new states that she didn't realize were possible.

Or she might learn of new outcomes that she didn't realize were available.

Or she might learn of new moral theories.

How should a rational agent update beliefs when her *awareness* changes?

How should she make decisions when she *knows* that she might be unaware of certain states and outcomes?

These are topics of ongoing research.

Fanaticism

SEU-maximization and all of its generalizations are vulnerable to *fanaticism*.

This refers to the Saint Petersburg Paradox, Pascal's Wager, and other choice problems where it seems "rational" to incur arbitrarily large costs for an infinitesimal chance of an astronomically large gain.

For an individual human, we can eliminate this problem by simply stipulating that the utility function is *bounded*.

But if we are deciding for a society (e.g. humanity) whose population (including future generations) could be arbitrarily large, then bounded utility is not only implausible, but leads to counterintuitive consequences.

Question. What theory of rational decisions avoids "fanatical" conclusions?

This is another topic of ongoing research.

Summary

We have reviewed a variety of theories for how to make rational decisions under uncertainty.

We have also reviewed a variety of theories for how to make moral decisions under moral uncertainty.

Some of these theories involve maximizing "expected choiceworthiness", so they can be seen as applications of the theory of rational decision-making.

However, because choiceworthiness is not necessarily cardinal and interpersonally comparable like utility, some other theories draw on methods from social choice theory.

Both decision theory and moral uncertainty are still areas of active research with many open problems.

Further reading

On decision theory

- *Prospect theory for risk and ambiguity*, by Peter Wakker (Cambridge University Press, 2010)
- *Theory of Decision under Uncertainty*, by Itzhak Gilboa (Cambridge University Press, 2009)
- *Rational Choice*, by Itzhak Gilboa (MIT Press, 2010)
- *An Introduction to Decision Theory*, by Martin Peterson (2nd edition, Cambridge University Press, 2017)
- "Decision Theory", by Katie Steele and H. Orri Stefánsson, *The Stanford Encyclopedia of Philosophy* (Fall 2025 Edition), Edward N. Zalta & Uri Nodelman (eds.),

On moral uncertainty

- *Moral Uncertainty*, by William MacAskill, Krister Bykvist and Toby Ord (Oxford University Press, 2020)
- "Book review of *Moral Uncertainty*," by M. Pivato, *Economics and Philosophy* **38**(#1), pp. 152-158 (2022)
- "Moral Decision-Making Under Uncertainty", by Christian Tarsney, Teruji Thomas, and William MacAskill, *The Stanford Encyclopedia of Philosophy* (Spring 2024 Edition), Edward N. Zalta & Uri Nodelman (eds.),